NC STATE UNIVERSITY

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CSC 525

Poster 36

Robust and Reliable Neural Networks: Analyzing Model Performance and Confidence under Noise and Adversarial Perturbations

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Problem Statement

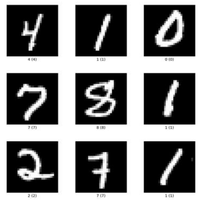
How reliable are neural network predictions under:
- Clean data
- Noisy conditions
- Adversarial attacks

Question: Can calibration methods improve trustworthiness?

Data

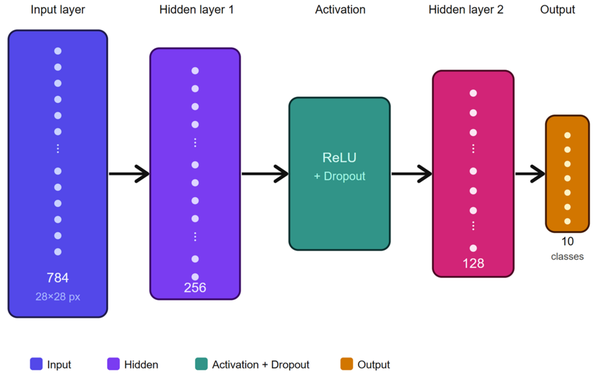
Dataset: MNIST
- 28×28 grayscale images
- 10 classes (digits 0–9)
- 60k train / 10k test

Preprocessing:
- Normalization (mean=0.1307, std=0.3081)

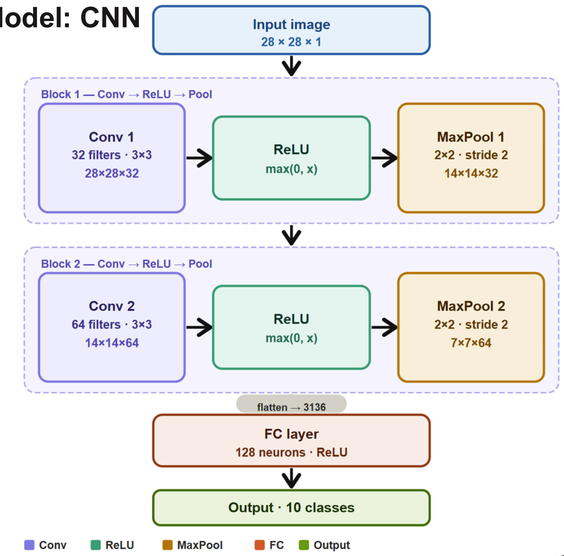


Network Architecture

Baseline: MLP



Advanced Model: CNN



Approach / Experimental Setup

1. Train MLP and CNN models
2. Evaluate on clean data
3. Add Gaussian noise (σ = 0.2)
4. Apply FGSM adversarial attack (ε = 0.15)
5. Measure:
   - Accuracy
   - F1
   - confidence
   - ECE
6. Apply temperature scaling for calibration

Evaluation Plan and Results

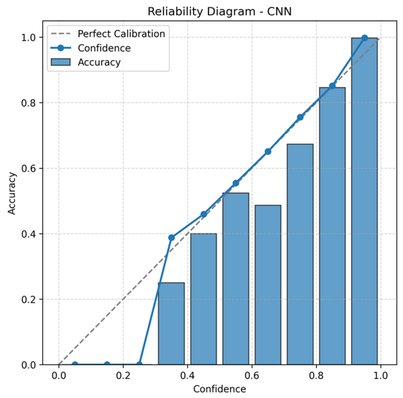
Evaluation metrics

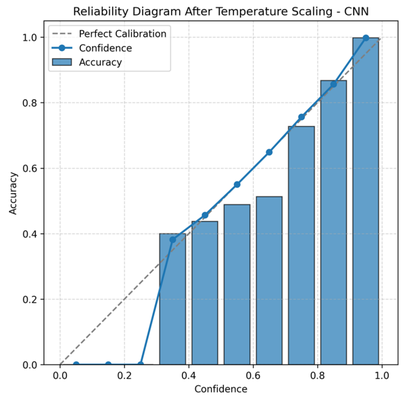
Model	Setting	Accuracy (%)	Macro F1 (%)	Avg Confidence (%)	Wrong Confidence (%)	ECE (%)	Insight
MLP	Clean	97.89	97.87	98.11	71.91	0.25	High accuracy, well-calibrated
	Noise	95.78	95.78	89.79	53.54	6.01	Performance drops, calibration worsens
	FGSM	43.97	43.18	57.25	54.50	13.97	Model collapses, highly unreliable
CNN	Clean	**99.10**	**99.09**	99.33	75.66	0.26	Excellent performance & calibration
	Noise	**99.03**	**99.02**	99.22	72.78	0.22	Highly robust to noise, CNN remains stable
	FGSM	**96.52**	**96.50**	98.09	81.16	1.65	Strong accuracy but overconfident

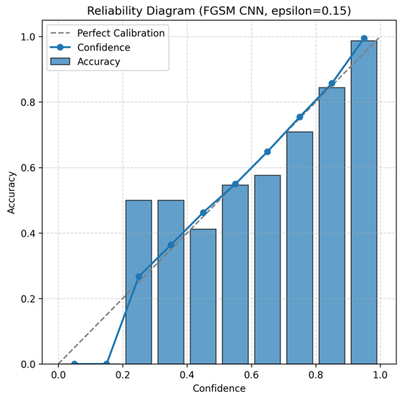
Temperature Scaling Evaluation

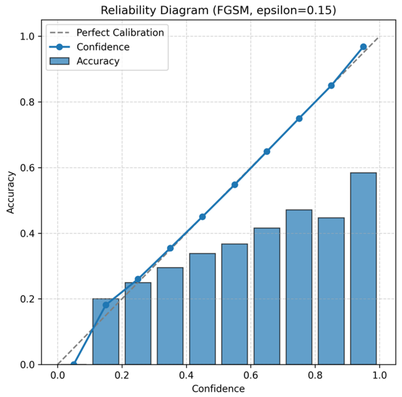
Model	Setting	Temperature (T)	Accuracy Before (%)	Accuracy After (%)	ECE Before (%)	ECE After (%)
MLP	CLEAN	1.0498	97.89	97.89	0.25	0.21
CNN		1.1356	99.10	99.10	0.26	0.18
MLP	NOISE	1.0866	96.19	96.19	4.30	5.61
CNN		1.1356	98.88	98.88	8.72	12.12
MLP	FGSM	1.0866	40.88	40.88	21.92	19.39
CNN		1.1356	92.87	92.87	11.32	15.30

Reliability Plots









Key Findings

- High accuracy ≠ reliable confidence
- CNN significantly outperforms MLP in robustness
- Models become overconfident under perturbations
- Adversarial attacks break reliability
- Calibration fails under adversarial settings

Main References

[1] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-Based Learning Applied to Document Recognition," Proceedings of the IEEE, vol. 86, no. 11, 1998.

[2] I. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and Harnessing Adversarial Examples," International Conference on Learning Representations (ICLR), 2015.

[3] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," International Conference on Machine Learning (ICML), 2017.

[4] PyTorch Vision, "MNIST Dataset," https://pytorch.org/vision/stable/datasets.html#mnist